REDESIGNING THE FRONT LINE: A DESIGN THINKING APPROACH TO EMERGENCY TRIAGE

AI IN HEALTHCARE

1 UNDERSTAND: The Friction

The Problem

- High rates of mis-triage, including both under-triage and over-triage, leading to delayed care or unnecessary resource use.
- Reported unsafe or inaccurate triage decisions in some primary care and telephone systems are around 10–20%.
- Heavy reliance on subjective clinical judgement leads to variability in decision-making between clinicians and settings, reducing consistency and reproducibility of triage decisions.
- Emergency departments operate under high cognitive load, where stress, fatigue, and time pressure reduce decision accuracy.
- Triage is often performed with incomplete patient information, especially in telephone triage, increasing uncertainty in urgency classification.
- Growing patient demand is placing increasing strain on emergency services and worsening bottlenecks, forcing faster decisions with fewer resources and compounding triage errors.
- Inconsistent adherence to triage protocols leads to guidelines being variably interpreted or bypassed in practice.
- Limited systematic feedback on triage decisions restricts opportunities for continuous improvement.

Citations: (NCBI, 2020; J. Clin. Epi, 2009).

Factors

Human Factors

- Cognitive load, stress/fatigue, subjective judgement, variability between clinicians.

Information Factors

- Incomplete patient info, limited feedback loops, inconsistent protocol application.

System Factors

- Growing demand, bottlenecks, 10-20% error rate.

User Affected

The NHS 111 call handler / primary care triage clinician, working under time pressure with incomplete information and no feedback on whether their decisions were right.

User

- An NHS triage nurse under time pressure.

Needs a way to

- Make consistent triage decisions with incomplete information.

Because

- 10-20% of decisions are unsafe, and they never find out which ones were wrong.

Key Insight

- The absence of a feedback loop is what makes the variability self-perpetuating.

2 DEFINE: Reframing the Challenge

Point of View Statement

- **User:** A telephone triage clinician (NHS 111) on a high-volume night shift.
- **Need:** A way to accurately detect "High-Risk/Low-Acuity" cases (e.g. early sepsis) amdist a sea of minor ailments.
- **Insight**: Mis-triage is a signal-detection problem exacerbated by cognitive fatigue, not a lack of clinical knowledge.

"How Might We" (HMW) Questions

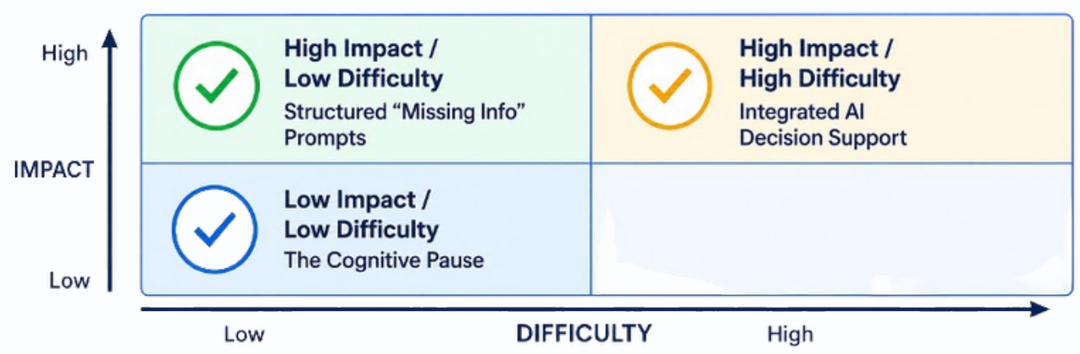
- **HMW** give staff a real-time "second opinion" for borderline cases?
- **HMW** make protocol adherence the "path of least resistance" in the software?
- **HMW** surface high-value missing data before a final urgency classification is made?

3 IDEATE: Comparison & Selection

Candidate Solutions

1. AI-Assisted Decision Support (Real-time flagging)
2. The "30-Second Cognitive Pause" protocol.
3. Mandatory "Missing Info" prompts in UI.
4. 24-hour Peer-Review "Buddy System".
5. Individualized Outcome-Feedback Dashboards.

The Decision Tool: Impact vs Difficulty Matrix



4 PROPOSE: The Recommended Solution

Solution Name

The Signal Booster

How It Works

- **Dynamic Checklists:** Forces the entry of "Red Flag" variables (vitals/symptoms) before a triage level is assigned
- **Working Memory Relief**: Acts as an external cognitive aid, reducing the "mental load" on the clinician.
- **Closing the Loop**: Automatically triggers a feedback notifaction to the clinician 48 hours later with the patient's actual hospital outcome.

The Value Prop

Reduces the 10-20% error rate by ensuring decisions are based on complete data, not just "gut feeling" during fatigue

Our Service

Data Insight
Human Decisions
Reduce A&E Waiting Times

Our Customer

We will target the ~£20+ billion per year public sector NHS Procurement Market.

Team Members and Github Repo

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https://github.com/salim-nader/Professional-Skils-Coursework-4-Group-8-.git